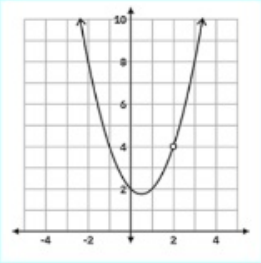
LIM-1.C.5 Numerical information can be used to estimate limits.

1-4 Estimating Limit Values from Tables

Given $f(x) = \{x^2 - x + 2 \quad \text{if } x \neq 2$

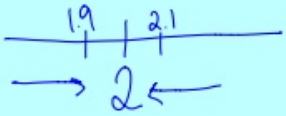


Use the graph to find the value of $\lim_{x \to 2} f(x)$: 4

How do we find the limit if we can't make a graph? We make a table. But just like a graph, we have to look at what happens when x approaches 2 from the left and when x approaches 2 from the right.

x	f(x)	x	f(x)
1.9	3.71	2.1	4.31
1.99	3.9701	2.01	4.0301
1.999	3.997001	2.001	4.003001
1.9999	3.99970001	2.0001	4.00030001
1.99999	3.9999700001	2.00001	4.0000300001

So: $\lim_{x \to 2} f(x)$: 4



Intuitive Definition of a Limit:

Suppose $f(x)$ is defined when x is near the number a. (This means that f is defined on some open interval that contains a, except possibly a itself.) Then we write:

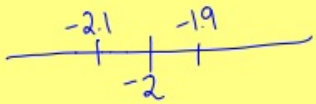
$$\lim_{x \to a} f(x) = \mathrm{L}$$

$$\text{if and only if } \lim_{x \to a^-} f(x) = \mathrm{L} \text{ and } \lim_{x \to a^+} f(x) = \mathrm{L}$$

In other words, in order for a limit to exist, the left- and right-hand limits must be equal. In that case the limit of the function equals the same value.

Example 1: If $f(x) = \frac{x^3-4x^2-7x+10}{x+2}$, create a table of values to evaluate $\lim_{x\to -2} f(x) =$ 21

x	-2.1	-2.01	-2.001	-1.999	-1.99	-1.9
$f(x)$	22.01	21.1001	21.010001	20.990001	20.9001	20.01



Example 2: The function f is continuous and increasing for $x \geq 1$. The table gives values of f at selected values of x. Approximate the value of: $\lim_{x \to 2} \cos\left(f(x)\right) = \cos\left(\lim_{x \to 2} f(x)\right) = \cos 5$

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	4.71	4.85	4.999	5.001	5.15	5.25

$\lim_{x \to 2} f(x) = 5$